Project Checkpoint Part 3: Statistical Modeling

Graduate students only (BIOL 5860)

PUT YOUR NAME HERE

This is the third of four checkpoint deliverables for the graduate project. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas by the end of Week 13. See the syllabus for the late policy.

1. Plan

Restate your hypothesis here.

Revisit your plan from Project Checkpoint Part 1 — update it here if your thinking has changed. Explain why any changes were made.

How will you test your stated hypothesis?

What is your alternative analysis plan (i.e. what will you do if your initial plan is not possible)?

2. Model

Based on the analysis plan, fit the statistical model or test that matches your design.

3. Assumptions and diagnostics

Check the assumptions relevant to your model (residual diagnostics, equal variance, overdispersion, or whatever applies) the way we've practiced throughout the course.

Do the diagnostics look acceptable? If not, add additional chunks below and walk through additional steps (transformation, restructuring, a nonparametric alternative, etc.). If you change your model, make sure you stick to your analysis plan to avoid p-hacking or HARKing.

3. Results

Report your results in a results-section style summary: model summary, effect sizes, test statistics, degrees of freedom, p-values, and confidence intervals where relevant. Where possible, use `marginaleffects` (`avg_comparisons()`, `avg_slopes()`, `avg_predictions()`) to report effect sizes in the units of your data.

4. Interpretation

In 3–5 sentences, interpret your results in the context of your original research question from Part 1. What do they mean biologically, and do they support or refute your hypotheses?